# Table 15 Reduced Form Effects on WHO Anthropometric Outcomes for Ages 0-5

WHO 2006 covers months 0-60 and WHO 2007 covers months 61 onward. Weight-for-length/height and related classifications are available through month 60 only. Father RDD estimates are the primary analysis and mother RDD estimates are secondary.

### Father RDD primary analysis

*Specification 1 Monthly child-age fixed effects  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.630 | 0.195 [0.175] N=2,374 | 0.176 [0.186] N=2,201 | 0.159 [0.200] N=1,983 | 0.327 [0.204] N=1,706 | 0.191 [0.234] N=1,410 |
| WHO weight-for-age z score | -0.115 | -0.051 [0.104] N=2,653 | -0.045 [0.114] N=2,459 | -0.017 [0.117] N=2,217 | -0.037 [0.131] N=1,903 | -0.093 [0.136] N=1,569 |
| WHO weight-for-length/height z score, 0-60 months | 0.329 | 0.086 [0.155] N=1,994 | 0.044 [0.159] N=1,853 | -0.006 [0.170] N=1,668 | -0.153 [0.185] N=1,430 | -0.057 [0.200] N=1,171 |
| WHO BMI-for-age z score | 0.348 | 0.027 [0.166] N=2,301 | 0.011 [0.169] N=2,137 | 0.026 [0.184] N=1,920 | -0.196 [0.191] N=1,651 | -0.096 [0.212] N=1,363 |
| Stunted: HAZ below -2 | 0.268 P=621 | -0.042 [0.033] N=2,374 | -0.047 [0.035] N=2,201 | -0.042 [0.036] N=1,983 | **-0.079** [0.037] N=1,706** | -0.036 [0.042] N=1,410 |
| Underweight: WAZ below -2 | 0.090 P=231 | 0.027 [0.021] N=2,653 | 0.020 [0.022] N=2,459 | 0.004 [0.024] N=2,217 | 0.007 [0.025] N=1,903 | 0.016 [0.029] N=1,569 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=227 | -0.034 [0.028] N=1,994 | -0.025 [0.028] N=1,853 | -0.018 [0.031] N=1,668 | -0.011 [0.034] N=1,430 | -0.034 [0.035] N=1,171 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.195 P=382 | 0.016 [0.038] N=1,994 | 0.008 [0.040] N=1,853 | 0.003 [0.041] N=1,668 | -0.020 [0.046] N=1,430 | -0.031 [0.051] N=1,171 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.089 P=179 | -0.014 [0.027] N=1,994 | -0.017 [0.029] N=1,853 | -0.005 [0.031] N=1,668 | -0.015 [0.034] N=1,430 | -0.018 [0.040] N=1,171 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.363 P=694 | **-0.063* [0.035] N=1,935** | **-0.067* [0.037] N=1,798** | -0.055 [0.038] N=1,618 | **-0.085** [0.041] N=1,388** | -0.064 [0.043] N=1,139 |
| BMI-for-age z score below -2 | 0.124 P=287 | -0.021 [0.028] N=2,301 | -0.023 [0.028] N=2,137 | -0.020 [0.030] N=1,920 | -0.002 [0.032] N=1,651 | -0.008 [0.035] N=1,363 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.410 P=128 | -0.089 [0.106] N=308 | -0.102 [0.108] N=286 | -0.059 [0.114] N=258 | -0.158 [0.121] N=228 | -0.179 [0.147] N=193 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.250 P=77 | -0.080 [0.089] N=308 | -0.128 [0.090] N=286 | -0.069 [0.098] N=258 | **-0.177* [0.105] N=228** | -0.176 [0.130] N=193 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=32 | 0.008 [0.061] N=308 | -0.036 [0.058] N=286 | -0.079 [0.061] N=258 | -0.064 [0.068] N=228 | **-0.146* [0.083] N=193** |

*Specification 1 Monthly child-age fixed effects  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.036 [0.244] N=1,998 | -0.045 [0.268] N=1,825 | -0.102 [0.308] N=1,607 | 0.190 [0.319] N=1,330 | -0.195 [0.438] N=1,034 |
| WHO weight-for-age z score | -0.194 [0.145] N=2,240 | -0.205 [0.165] N=2,046 | -0.178 [0.176] N=1,804 | -0.279 [0.216] N=1,490 | **-0.479* [0.246] N=1,156** |
| WHO weight-for-length/height z score, 0-60 months | 0.106 [0.230] N=1,680 | 0.042 [0.245] N=1,539 | -0.073 [0.276] N=1,354 | -0.382 [0.339] N=1,116 | -0.283 [0.425] N=857 |
| WHO BMI-for-age z score | 0.084 [0.248] N=1,934 | 0.082 [0.261] N=1,770 | 0.124 [0.307] N=1,553 | -0.311 [0.356] N=1,284 | -0.120 [0.453] N=996 |
| Stunted: HAZ below -2 | -0.018 [0.046] N=1,998 | -0.021 [0.051] N=1,825 | -0.014 [0.054] N=1,607 | -0.083 [0.061] N=1,330 | 0.021 [0.081] N=1,034 |
| Underweight: WAZ below -2 | 0.041 [0.029] N=2,240 | 0.029 [0.033] N=2,046 | -0.001 [0.036] N=1,804 | 0.010 [0.042] N=1,490 | 0.038 [0.057] N=1,156 |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.032 [0.038] N=1,680 | -0.017 [0.039] N=1,539 | -0.003 [0.045] N=1,354 | 0.012 [0.055] N=1,116 | -0.040 [0.064] N=857 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.025 [0.045] N=1,680 | 0.014 [0.050] N=1,539 | 0.001 [0.052] N=1,354 | -0.036 [0.066] N=1,116 | -0.097 [0.084] N=857 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.022 [0.031] N=1,680 | -0.029 [0.034] N=1,539 | -0.007 [0.036] N=1,354 | -0.026 [0.044] N=1,116 | -0.047 [0.062] N=857 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.039 [0.048] N=1,632 | -0.043 [0.055] N=1,495 | -0.027 [0.059] N=1,315 | -0.084 [0.069] N=1,085 | -0.044 [0.083] N=836 |
| BMI-for-age z score below -2 | -0.038 [0.038] N=1,934 | -0.050 [0.038] N=1,770 | -0.053 [0.044] N=1,553 | -0.030 [0.051] N=1,284 | -0.074 [0.066] N=996 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.048 [0.161] N=254 | -0.038 [0.171] N=232 | 0.064 [0.186] N=204 | -0.118 [0.208] N=174 | -0.176 [0.285] N=139 |
| WHO 2007 BMI obesity above +2, 61+ months | -0.011 [0.122] N=254 | -0.076 [0.130] N=232 | 0.042 [0.153] N=204 | -0.160 [0.177] N=174 | -0.157 [0.242] N=139 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.051 [0.094] N=254 | -0.037 [0.091] N=232 | -0.112 [0.100] N=204 | -0.092 [0.121] N=174 | **-0.326** [0.155] N=139** |

*Specification 2 Monthly fixed effects and female  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.630 | 0.195 [0.176] N=2,374 | 0.176 [0.186] N=2,201 | 0.162 [0.200] N=1,983 | 0.332 [0.203] N=1,706 | 0.196 [0.233] N=1,410 |
| WHO weight-for-age z score | -0.115 | -0.053 [0.104] N=2,653 | -0.047 [0.113] N=2,459 | -0.019 [0.117] N=2,217 | -0.038 [0.130] N=1,903 | -0.096 [0.135] N=1,569 |
| WHO weight-for-length/height z score, 0-60 months | 0.329 | 0.085 [0.155] N=1,994 | 0.042 [0.160] N=1,853 | -0.010 [0.171] N=1,668 | -0.157 [0.185] N=1,430 | -0.068 [0.200] N=1,171 |
| WHO BMI-for-age z score | 0.348 | 0.030 [0.167] N=2,301 | 0.014 [0.170] N=2,137 | 0.027 [0.185] N=1,920 | -0.194 [0.191] N=1,651 | -0.098 [0.212] N=1,363 |
| Stunted: HAZ below -2 | 0.268 P=621 | -0.041 [0.033] N=2,374 | -0.046 [0.035] N=2,201 | -0.041 [0.036] N=1,983 | **-0.078** [0.037] N=1,706** | -0.036 [0.042] N=1,410 |
| Underweight: WAZ below -2 | 0.090 P=231 | 0.028 [0.021] N=2,653 | 0.020 [0.022] N=2,459 | 0.004 [0.024] N=2,217 | 0.008 [0.025] N=1,903 | 0.016 [0.028] N=1,569 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=227 | -0.032 [0.027] N=1,994 | -0.022 [0.028] N=1,853 | -0.014 [0.030] N=1,668 | -0.007 [0.033] N=1,430 | -0.029 [0.035] N=1,171 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.195 P=382 | 0.018 [0.037] N=1,994 | 0.010 [0.039] N=1,853 | 0.005 [0.041] N=1,668 | -0.017 [0.046] N=1,430 | -0.028 [0.051] N=1,171 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.089 P=179 | -0.013 [0.027] N=1,994 | -0.015 [0.029] N=1,853 | -0.003 [0.031] N=1,668 | -0.012 [0.034] N=1,430 | -0.015 [0.040] N=1,171 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.363 P=694 | **-0.060* [0.034] N=1,935** | **-0.063* [0.037] N=1,798** | -0.050 [0.037] N=1,618 | **-0.079** [0.040] N=1,388** | -0.056 [0.042] N=1,139 |
| BMI-for-age z score below -2 | 0.124 P=287 | -0.021 [0.028] N=2,301 | -0.022 [0.028] N=2,137 | -0.019 [0.030] N=1,920 | -0.001 [0.032] N=1,651 | -0.007 [0.035] N=1,363 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.410 P=128 | -0.092 [0.105] N=308 | -0.109 [0.108] N=286 | -0.067 [0.115] N=258 | -0.181 [0.120] N=228 | -0.199 [0.144] N=193 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.250 P=77 | -0.082 [0.089] N=308 | -0.134 [0.090] N=286 | -0.078 [0.099] N=258 | **-0.199* [0.107] N=228** | -0.196 [0.129] N=193 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=32 | 0.013 [0.061] N=308 | -0.028 [0.059] N=286 | -0.068 [0.061] N=258 | -0.047 [0.067] N=228 | -0.129 [0.082] N=193 |

*Specification 2 Monthly fixed effects and female 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.034 [0.244] N=1,998 | -0.046 [0.268] N=1,825 | -0.104 [0.307] N=1,607 | 0.184 [0.320] N=1,330 | -0.219 [0.441] N=1,034 |
| WHO weight-for-age z score | -0.194 [0.145] N=2,240 | -0.205 [0.165] N=2,046 | -0.178 [0.176] N=1,804 | -0.279 [0.216] N=1,490 | **-0.477* [0.246] N=1,156** |
| WHO weight-for-length/height z score, 0-60 months | 0.106 [0.230] N=1,680 | 0.040 [0.245] N=1,539 | -0.077 [0.276] N=1,354 | -0.387 [0.339] N=1,116 | -0.288 [0.424] N=857 |
| WHO BMI-for-age z score | 0.083 [0.249] N=1,934 | 0.082 [0.261] N=1,770 | 0.124 [0.307] N=1,553 | -0.311 [0.356] N=1,284 | -0.113 [0.453] N=996 |
| Stunted: HAZ below -2 | -0.018 [0.046] N=1,998 | -0.021 [0.051] N=1,825 | -0.014 [0.054] N=1,607 | -0.083 [0.061] N=1,330 | 0.023 [0.082] N=1,034 |
| Underweight: WAZ below -2 | 0.040 [0.029] N=2,240 | 0.029 [0.033] N=2,046 | -0.001 [0.036] N=1,804 | 0.010 [0.042] N=1,490 | 0.037 [0.057] N=1,156 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.032 [0.038] N=1,680 | -0.016 [0.039] N=1,539 | -0.002 [0.046] N=1,354 | 0.014 [0.056] N=1,116 | -0.039 [0.064] N=857 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.025 [0.045] N=1,680 | 0.015 [0.050] N=1,539 | 0.002 [0.052] N=1,354 | -0.034 [0.066] N=1,116 | -0.096 [0.084] N=857 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.022 [0.031] N=1,680 | -0.029 [0.035] N=1,539 | -0.006 [0.036] N=1,354 | -0.024 [0.044] N=1,116 | -0.047 [0.063] N=857 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.040 [0.048] N=1,632 | -0.043 [0.055] N=1,495 | -0.027 [0.058] N=1,315 | -0.083 [0.069] N=1,085 | -0.044 [0.083] N=836 |
| BMI-for-age z score below -2 | -0.038 [0.038] N=1,934 | -0.050 [0.038] N=1,770 | -0.053 [0.044] N=1,553 | -0.031 [0.051] N=1,284 | -0.075 [0.066] N=996 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.048 [0.162] N=254 | -0.042 [0.173] N=232 | 0.062 [0.189] N=204 | -0.144 [0.215] N=174 | -0.181 [0.292] N=139 |
| WHO 2007 BMI obesity above +2, 61+ months | -0.010 [0.120] N=254 | -0.082 [0.129] N=232 | 0.030 [0.154] N=204 | -0.203 [0.176] N=174 | -0.190 [0.241] N=139 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.050 [0.093] N=254 | -0.025 [0.091] N=232 | -0.092 [0.099] N=204 | -0.053 [0.116] N=174 | **-0.283* [0.152] N=139** |

*Specification 3 Monthly fixed effects female and household income groups No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.630 | 0.170 [0.176] N=2,374 | 0.146 [0.187] N=2,201 | 0.132 [0.202] N=1,983 | 0.297 [0.205] N=1,706 | 0.152 [0.235] N=1,410 |
| WHO weight-for-age z score | -0.115 | -0.086 [0.101] N=2,653 | -0.080 [0.111] N=2,459 | -0.054 [0.115] N=2,217 | -0.070 [0.128] N=1,903 | -0.135 [0.134] N=1,569 |
| WHO weight-for-length/height z score, 0-60 months | 0.329 | 0.077 [0.158] N=1,994 | 0.039 [0.164] N=1,853 | -0.008 [0.174] N=1,668 | -0.155 [0.187] N=1,430 | -0.069 [0.203] N=1,171 |
| WHO BMI-for-age z score | 0.348 | 0.029 [0.168] N=2,301 | 0.016 [0.171] N=2,137 | 0.032 [0.186] N=1,920 | -0.183 [0.192] N=1,651 | -0.084 [0.215] N=1,363 |
| Stunted: HAZ below -2 | 0.268 P=621 | -0.034 [0.032] N=2,374 | -0.036 [0.034] N=2,201 | -0.032 [0.035] N=1,983 | **-0.067* [0.037] N=1,706** | -0.022 [0.041] N=1,410 |
| Underweight: WAZ below -2 | 0.090 P=231 | **0.035* [0.020] N=2,653** | 0.028 [0.022] N=2,459 | 0.013 [0.023] N=2,217 | 0.017 [0.024] N=1,903 | 0.028 [0.027] N=1,569 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=227 | -0.028 [0.027] N=1,994 | -0.017 [0.028] N=1,853 | -0.011 [0.031] N=1,668 | -0.005 [0.033] N=1,430 | -0.026 [0.035] N=1,171 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.195 P=382 | 0.021 [0.038] N=1,994 | 0.014 [0.040] N=1,853 | 0.009 [0.041] N=1,668 | -0.014 [0.046] N=1,430 | -0.024 [0.051] N=1,171 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.089 P=179 | -0.012 [0.027] N=1,994 | -0.013 [0.029] N=1,853 | -0.001 [0.031] N=1,668 | -0.011 [0.034] N=1,430 | -0.015 [0.040] N=1,171 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.363 P=694 | -0.046 [0.033] N=1,935 | -0.047 [0.035] N=1,798 | -0.039 [0.036] N=1,618 | **-0.071* [0.038] N=1,388** | -0.041 [0.041] N=1,139 |

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| BMI-for-age z score below -2 | 0.124 P=287 | -0.017 [0.028] N=2,301 | -0.018 [0.028] N=2,137 | -0.016 [0.031] N=1,920 | 0.002 [0.032] N=1,651 | -0.004 [0.036] N=1,363 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.410 P=128 | -0.087 [0.106] N=308 | -0.097 [0.108] N=286 | -0.050 [0.113] N=258 | -0.166 [0.119] N=228 | -0.186 [0.147] N=193 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.250 P=77 | -0.073 [0.089] N=308 | -0.118 [0.091] N=286 | -0.055 [0.099] N=258 | **-0.175* [0.103] N=228** | -0.170 [0.128] N=193 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=32 | 0.016 [0.059] N=308 | -0.017 [0.057] N=286 | -0.056 [0.059] N=258 | -0.022 [0.065] N=228 | -0.106 [0.081] N=193 |

*Specification 3 Monthly fixed effects female and household income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.017 [0.243] N=1,998 | -0.077 [0.266] N=1,825 | -0.137 [0.304] N=1,607 | 0.123 [0.317] N=1,330 | -0.282 [0.435] N=1,034 |
| WHO weight-for-age z score | -0.217 [0.141] N=2,240 | -0.231 [0.163] N=2,046 | -0.204 [0.172] N=1,804 | -0.298 [0.212] N=1,490 | **-0.507** [0.243] N=1,156** |
| WHO weight-for-length/height z score, 0-60 months | 0.100 [0.232] N=1,680 | 0.041 [0.251] N=1,539 | -0.068 [0.277] N=1,354 | -0.393 [0.338] N=1,116 | -0.314 [0.425] N=857 |
| WHO BMI-for-age z score | 0.087 [0.250] N=1,934 | 0.090 [0.264] N=1,770 | 0.140 [0.307] N=1,553 | -0.283 [0.356] N=1,284 | -0.078 [0.453] N=996 |
| Stunted: HAZ below -2 | -0.014 [0.045] N=1,998 | -0.013 [0.050] N=1,825 | -0.007 [0.053] N=1,607 | -0.070 [0.059] N=1,330 | 0.039 [0.078] N=1,034 |
| Underweight: WAZ below -2 | 0.045 [0.028] N=2,240 | 0.036 [0.032] N=2,046 | 0.005 [0.035] N=1,804 | 0.014 [0.041] N=1,490 | 0.045 [0.056] N=1,156 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.028 [0.038] N=1,680 | -0.010 [0.040] N=1,539 | 0.001 [0.046] N=1,354 | 0.017 [0.057] N=1,116 | -0.033 [0.066] N=857 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.028 [0.046] N=1,680 | 0.020 [0.051] N=1,539 | 0.004 [0.053] N=1,354 | -0.035 [0.066] N=1,116 | -0.091 [0.084] N=857 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.022 [0.031] N=1,680 | -0.026 [0.035] N=1,539 | -0.006 [0.036] N=1,354 | -0.026 [0.044] N=1,116 | -0.052 [0.062] N=857 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.027 [0.046] N=1,632 | -0.026 [0.052] N=1,495 | -0.018 [0.055] N=1,315 | -0.081 [0.065] N=1,085 | -0.034 [0.078] N=836 |
| BMI-for-age z score below -2 | -0.035 [0.038] N=1,934 | -0.045 [0.038] N=1,770 | -0.051 [0.045] N=1,553 | -0.031 [0.052] N=1,284 | -0.074 [0.067] N=996 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.051 [0.158] N=254 | -0.044 [0.171] N=232 | 0.070 [0.183] N=204 | -0.132 [0.210] N=174 | -0.172 [0.292] N=139 |
| WHO 2007 BMI obesity above +2, 61+ months | -0.016 [0.118] N=254 | -0.077 [0.129] N=232 | 0.044 [0.151] N=204 | -0.181 [0.171] N=174 | -0.184 [0.240] N=139 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.056 [0.090] N=254 | -0.016 [0.087] N=232 | -0.087 [0.096] N=204 | -0.013 [0.110] N=174 | -0.212 [0.150] N=139 |

*Specification 4 Monthly fixed effects female and per-person income groups No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.630 | 0.150 [0.175] N=2,374 | 0.134 [0.186] N=2,201 | 0.118 [0.201] N=1,983 | 0.275 [0.204] N=1,706 | 0.125 [0.233] N=1,410 |
| WHO weight-for-age z score | -0.115 | -0.089 [0.102] N=2,653 | -0.091 [0.111] N=2,459 | -0.067 [0.116] N=2,217 | -0.088 [0.130] N=1,903 | -0.170 [0.137] N=1,569 |
| WHO weight-for-length/height z score, 0-60 months | 0.329 | 0.090 [0.157] N=1,994 | 0.044 [0.162] N=1,853 | -0.006 [0.173] N=1,668 | -0.152 [0.187] N=1,430 | -0.072 [0.203] N=1,171 |
| WHO BMI-for-age z score | 0.348 | 0.041 [0.169] N=2,301 | 0.022 [0.171] N=2,137 | 0.039 [0.186] N=1,920 | -0.173 [0.194] N=1,651 | -0.080 [0.217] N=1,363 |

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Stunted: HAZ below -2 | 0.268 P=621 | -0.031 [0.032] N=2,374 | -0.035 [0.034] N=2,201 | -0.030 [0.035] N=1,983 | **-0.063* [0.036] N=1,706** | -0.019 [0.040] N=1,410 |
| Underweight: WAZ below -2 | 0.090 P=231 | **0.035* [0.020] N=2,653** | 0.030 [0.022] N=2,459 | 0.016 [0.023] N=2,217 | 0.020 [0.024] N=1,903 | 0.033 [0.027] N=1,569 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=227 | -0.029 [0.027] N=1,994 | -0.018 [0.028] N=1,853 | -0.010 [0.030] N=1,668 | -0.005 [0.033] N=1,430 | -0.027 [0.035] N=1,171 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.195 P=382 | 0.024 [0.038] N=1,994 | 0.017 [0.040] N=1,853 | 0.011 [0.041] N=1,668 | -0.011 [0.046] N=1,430 | -0.025 [0.051] N=1,171 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.089 P=179 | -0.011 [0.027] N=1,994 | -0.015 [0.029] N=1,853 | -0.003 [0.030] N=1,668 | -0.011 [0.034] N=1,430 | -0.017 [0.040] N=1,171 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.363 P=694 | -0.040 [0.033] N=1,935 | -0.041 [0.035] N=1,798 | -0.031 [0.036] N=1,618 | -0.061 [0.038] N=1,388 | -0.033 [0.041] N=1,139 |
| BMI-for-age z score below -2 | 0.124 P=287 | -0.018 [0.028] N=2,301 | -0.019 [0.028] N=2,137 | -0.015 [0.031] N=1,920 | 0.004 [0.032] N=1,651 | -0.003 [0.036] N=1,363 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.410 P=128 | -0.086 [0.104] N=308 | -0.105 [0.108] N=286 | -0.058 [0.114] N=258 | -0.153 [0.119] N=228 | -0.169 [0.143] N=193 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.250 P=77 | -0.077 [0.089] N=308 | -0.128 [0.091] N=286 | -0.068 [0.099] N=258 | -0.179 [0.109] N=228 | -0.175 [0.136] N=193 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=32 | -0.004 [0.059] N=308 | -0.029 [0.058] N=286 | -0.080 [0.059] N=258 | -0.059 [0.064] N=228 | -0.123 [0.077] N=193 |

*Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.006 [0.241] N=1,998 | -0.075 [0.265] N=1,825 | -0.129 [0.303] N=1,607 | 0.159 [0.317] N=1,330 | -0.261 [0.433] N=1,034 |
| WHO weight-for-age z score | -0.208 [0.143] N=2,240 | -0.231 [0.163] N=2,046 | -0.209 [0.174] N=1,804 | -0.293 [0.215] N=1,490 | **-0.489* [0.247] N=1,156** |
| WHO weight-for-length/height z score, 0-60 months | 0.107 [0.232] N=1,680 | 0.044 [0.249] N=1,539 | -0.073 [0.280] N=1,354 | -0.392 [0.342] N=1,116 | -0.293 [0.426] N=857 |
| WHO BMI-for-age z score | 0.094 [0.250] N=1,934 | 0.089 [0.262] N=1,770 | 0.136 [0.309] N=1,553 | -0.289 [0.358] N=1,284 | -0.082 [0.455] N=996 |
| Stunted: HAZ below -2 | -0.012 [0.045] N=1,998 | -0.014 [0.050] N=1,825 | -0.009 [0.053] N=1,607 | -0.077 [0.059] N=1,330 | 0.037 [0.078] N=1,034 |
| Underweight: WAZ below -2 | 0.044 [0.028] N=2,240 | 0.035 [0.032] N=2,046 | 0.007 [0.035] N=1,804 | 0.015 [0.041] N=1,490 | 0.045 [0.056] N=1,156 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.027 [0.038] N=1,680 | -0.013 [0.040] N=1,539 | 0.000 [0.046] N=1,354 | 0.018 [0.057] N=1,116 | -0.033 [0.066] N=857 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.029 [0.046] N=1,680 | 0.021 [0.050] N=1,539 | 0.004 [0.053] N=1,354 | -0.035 [0.066] N=1,116 | -0.090 [0.084] N=857 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.019 [0.031] N=1,680 | -0.025 [0.035] N=1,539 | -0.004 [0.037] N=1,354 | -0.025 [0.044] N=1,116 | -0.052 [0.060] N=857 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.023 [0.047] N=1,632 | -0.027 [0.052] N=1,495 | -0.017 [0.056] N=1,315 | -0.082 [0.067] N=1,085 | -0.030 [0.081] N=836 |
| BMI-for-age z score below -2 | -0.035 [0.038] N=1,934 | -0.047 [0.038] N=1,770 | -0.052 [0.045] N=1,553 | -0.029 [0.052] N=1,284 | -0.072 [0.067] N=996 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.031 [0.159] N=254 | -0.040 [0.175] N=232 | 0.068 [0.187] N=204 | -0.116 [0.214] N=174 | -0.110 [0.301] N=139 |
| WHO 2007 BMI obesity above +2, 61+ months | -0.009 [0.118] N=254 | -0.082 [0.127] N=232 | 0.027 [0.147] N=204 | -0.191 [0.175] N=174 | -0.143 [0.256] N=139 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.029 [0.089] N=254 | -0.020 [0.089] N=232 | -0.108 [0.096] N=204 | -0.069 [0.111] N=174 | **-0.269* [0.144] N=139** |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.640 | 0.148 [0.187] N=2,220 | 0.124 [0.198] N=2,063 | 0.095 [0.213] N=1,854 | 0.257 [0.213] N=1,604 | 0.117 [0.237] N=1,323 |
| WHO weight-for-age z score | -0.113 | -0.025 [0.106] N=2,471 | -0.028 [0.114] N=2,295 | 0.000 [0.121] N=2,064 | -0.028 [0.134] N=1,780 | -0.069 [0.143] N=1,467 |
| WHO weight-for-length/height z score, 0-60 months | 0.338 | 0.193 [0.160] N=1,859 | 0.141 [0.163] N=1,735 | 0.099 [0.172] N=1,559 | -0.039 [0.190] N=1,346 | 0.096 [0.212] N=1,100 |
| WHO BMI-for-age z score | 0.357 | 0.125 [0.172] N=2,151 | 0.090 [0.174] N=2,002 | 0.124 [0.189] N=1,796 | -0.073 [0.197] N=1,554 | 0.059 [0.220] N=1,281 |
| Stunted: HAZ below -2 | 0.268 P=580 | -0.036 [0.036] N=2,220 | -0.041 [0.038] N=2,063 | -0.033 [0.039] N=1,854 | **-0.070* [0.041] N=1,604** | -0.024 [0.045] N=1,323 |
| Underweight: WAZ below -2 | 0.091 P=217 | 0.024 [0.021] N=2,471 | 0.019 [0.023] N=2,295 | 0.003 [0.025] N=2,064 | 0.009 [0.026] N=1,780 | 0.020 [0.030] N=1,467 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=211 | -0.043 [0.028] N=1,859 | -0.032 [0.028] N=1,735 | -0.020 [0.031] N=1,559 | -0.014 [0.034] N=1,346 | -0.035 [0.036] N=1,100 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.196 P=357 | 0.028 [0.039] N=1,859 | 0.019 [0.041] N=1,735 | 0.018 [0.043] N=1,559 | -0.006 [0.048] N=1,346 | -0.006 [0.056] N=1,100 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.091 P=172 | -0.013 [0.029] N=1,859 | -0.014 [0.031] N=1,735 | 0.004 [0.033] N=1,559 | -0.003 [0.036] N=1,346 | -0.001 [0.042] N=1,100 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.364 P=647 | -0.060 [0.039] N=1,805 | -0.062 [0.041] N=1,684 | -0.046 [0.042] N=1,512 | **-0.078* [0.046] N=1,307** | -0.048 [0.049] N=1,070 |
| BMI-for-age z score below -2 | 0.123 P=264 | -0.029 [0.029] N=2,151 | -0.029 [0.028] N=2,002 | -0.028 [0.031] N=1,796 | -0.012 [0.033] N=1,554 | -0.022 [0.037] N=1,281 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.406 P=120 | -0.111 [0.109] N=293 | -0.116 [0.116] N=271 | -0.073 [0.123] N=244 | -0.204 [0.124] N=216 | -0.194 [0.154] N=183 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.244 P=71 | -0.094 [0.093] N=293 | -0.135 [0.096] N=271 | -0.079 [0.107] N=244 | **-0.222* [0.112] N=216** | **-0.230* [0.138] N=183** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.106 P=29 | 0.016 [0.054] N=293 | -0.032 [0.053] N=271 | -0.082 [0.053] N=244 | -0.060 [0.058] N=216 | **-0.148** [0.066] N=183** |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.061 [0.265] N=1,869 | -0.016 [0.293] N=1,712 | -0.088 [0.335] N=1,503 | 0.216 [0.360] N=1,253 | -0.139 [0.488] N=972 |
| WHO weight-for-age z score | -0.181 [0.145] N=2,086 | -0.213 [0.161] N=1,910 | -0.200 [0.176] N=1,679 | -0.343 [0.213] N=1,395 | **-0.525** [0.250] N=1,082** |
| WHO weight-for-length/height z score, 0-60 months | 0.125 [0.239] N=1,565 | 0.018 [0.250] N=1,441 | -0.088 [0.280] N=1,265 | -0.383 [0.351] N=1,052 | -0.194 [0.441] N=806 |
| WHO BMI-for-age z score | 0.069 [0.257] N=1,808 | 0.020 [0.269] N=1,659 | 0.082 [0.315] N=1,453 | -0.330 [0.373] N=1,211 | -0.077 [0.467] N=938 |
| Stunted: HAZ below -2 | -0.027 [0.049] N=1,869 | -0.033 [0.055] N=1,712 | -0.024 [0.059] N=1,503 | -0.102 [0.068] N=1,253 | 0.002 [0.090] N=972 |
| Underweight: WAZ below -2 | 0.036 [0.030] N=2,086 | 0.028 [0.034] N=1,910 | 0.001 [0.038] N=1,679 | 0.021 [0.044] N=1,395 | 0.054 [0.059] N=1,082 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.048 [0.038] N=1,565 | -0.029 [0.039] N=1,441 | -0.010 [0.045] N=1,265 | -0.003 [0.056] N=1,052 | -0.066 [0.061] N=806 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.022 [0.047] N=1,565 | 0.004 [0.050] N=1,441 | -0.002 [0.052] N=1,265 | -0.045 [0.068] N=1,052 | -0.079 [0.089] N=806 |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Obese: WLZ/WHZ above +3, 0-60 months | -0.036 [0.033] N=1,565 | -0.043 [0.038] N=1,441 | -0.013 [0.039] N=1,265 | -0.028 [0.047] N=1,052 | -0.042 [0.066] N=806 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.053 [0.051] N=1,522 | -0.061 [0.057] N=1,401 | -0.040 [0.064] N=1,229 | -0.117 [0.075] N=1,024 | -0.077 [0.092] N=787 |
| BMI-for-age z score below -2 | -0.041 [0.039] N=1,808 | -0.050 [0.038] N=1,659 | -0.054 [0.045] N=1,453 | -0.041 [0.052] N=1,211 | -0.102 [0.066] N=938 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.086 [0.175] N=242 | -0.064 [0.193] N=220 | 0.045 [0.208] N=193 | -0.166 [0.227] N=165 | -0.177 [0.320] N=132 |
| WHO 2007 BMI obesity above +2, 61+ months | -0.042 [0.129] N=242 | -0.097 [0.140] N=220 | 0.004 [0.162] N=193 | -0.255 [0.181] N=165 | -0.303 [0.251] N=132 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.081 [0.097] N=242 | -0.006 [0.098] N=220 | -0.095 [0.109] N=193 | -0.067 [0.129] N=165 | **-0.334** [0.152] N=132** |

*Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.640 | 0.157 [0.185] N=2,220 | 0.132 [0.196] N=2,063 | 0.098 [0.212] N=1,854 | 0.261 [0.213] N=1,604 | 0.125 [0.237] N=1,323 |
| WHO weight-for-age z score | -0.113 | -0.016 [0.106] N=2,471 | -0.021 [0.115] N=2,295 | 0.002 [0.120] N=2,064 | -0.025 [0.134] N=1,780 | -0.067 [0.144] N=1,467 |
| WHO weight-for-length/height z score, 0-60 months | 0.338 | 0.193 [0.159] N=1,859 | 0.143 [0.162] N=1,735 | 0.091 [0.171] N=1,559 | -0.046 [0.190] N=1,346 | 0.072 [0.212] N=1,100 |
| WHO BMI-for-age z score | 0.357 | 0.127 [0.171] N=2,151 | 0.092 [0.173] N=2,002 | 0.124 [0.188] N=1,796 | -0.074 [0.197] N=1,554 | 0.052 [0.221] N=1,281 |
| Stunted: HAZ below -2 | 0.268 P=580 | -0.038 [0.035] N=2,220 | -0.043 [0.038] N=2,063 | -0.036 [0.039] N=1,854 | **-0.073* [0.040] N=1,604** | -0.030 [0.044] N=1,323 |
| Underweight: WAZ below -2 | 0.091 P=217 | 0.021 [0.021] N=2,471 | 0.016 [0.023] N=2,295 | 0.001 [0.024] N=2,064 | 0.008 [0.026] N=1,780 | 0.018 [0.030] N=1,467 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=211 | -0.043 [0.028] N=1,859 | -0.032 [0.028] N=1,735 | -0.021 [0.031] N=1,559 | -0.015 [0.034] N=1,346 | -0.037 [0.036] N=1,100 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.196 P=357 | 0.028 [0.039] N=1,859 | 0.019 [0.041] N=1,735 | 0.016 [0.043] N=1,559 | -0.010 [0.048] N=1,346 | -0.013 [0.055] N=1,100 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.091 P=172 | -0.013 [0.029] N=1,859 | -0.013 [0.031] N=1,735 | 0.003 [0.033] N=1,559 | -0.004 [0.036] N=1,346 | -0.004 [0.042] N=1,100 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.364 P=647 | -0.060 [0.038] N=1,805 | -0.062 [0.041] N=1,684 | -0.047 [0.041] N=1,512 | **-0.082* [0.045] N=1,307** | -0.057 [0.046] N=1,070 |
| BMI-for-age z score below -2 | 0.123 P=264 | -0.030 [0.029] N=2,151 | -0.030 [0.028] N=2,002 | -0.028 [0.031] N=1,796 | -0.012 [0.033] N=1,554 | -0.022 [0.037] N=1,281 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.406 P=120 | -0.103 [0.106] N=293 | -0.107 [0.112] N=271 | -0.067 [0.120] N=244 | -0.192 [0.123] N=216 | -0.201 [0.151] N=183 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.244 P=71 | -0.097 [0.090] N=293 | -0.140 [0.093] N=271 | -0.080 [0.104] N=244 | **-0.223** [0.110] N=216** | **-0.229* [0.137] N=183** |
| WHO 2007 BMI thinness below -2, 61+ months | 0.106 P=29 | 0.004 [0.054] N=293 | -0.042 [0.053] N=271 | -0.086 [0.053] N=244 | -0.065 [0.057] N=216 | **-0.155** [0.066] N=183** |

*Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | 0.074 [0.264] N=1,869 | -0.006 [0.292] N=1,712 | -0.083 [0.335] N=1,503 | 0.219 [0.362] N=1,253 | -0.106 [0.491] N=972 |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO weight-for-age z score | -0.163 [0.144] N=2,086 | -0.197 [0.160] N=1,910 | -0.190 [0.173] N=1,679 | -0.326 [0.211] N=1,395 | **-0.498** [0.249] N=1,082** |
| WHO weight-for-length/height z score, 0-60 months | 0.155 [0.238] N=1,565 | 0.061 [0.251] N=1,441 | -0.075 [0.281] N=1,265 | -0.372 [0.355] N=1,052 | -0.200 [0.450] N=806 |
| WHO BMI-for-age z score | 0.085 [0.255] N=1,808 | 0.038 [0.267] N=1,659 | 0.088 [0.314] N=1,453 | -0.323 [0.371] N=1,211 | -0.080 [0.470] N=938 |
| Stunted: HAZ below -2 | -0.029 [0.049] N=1,869 | -0.034 [0.055] N=1,712 | -0.027 [0.060] N=1,503 | -0.105 [0.068] N=1,253 | -0.010 [0.091] N=972 |
| Underweight: WAZ below -2 | 0.033 [0.030] N=2,086 | 0.025 [0.034] N=1,910 | -0.003 [0.038] N=1,679 | 0.016 [0.043] N=1,395 | 0.047 [0.059] N=1,082 |
| Wasted: WLZ/WHZ below -2, 0-60 months | -0.048 [0.038] N=1,565 | -0.031 [0.039] N=1,441 | -0.010 [0.046] N=1,265 | -0.005 [0.056] N=1,052 | -0.073 [0.062] N=806 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.031 [0.046] N=1,565 | 0.016 [0.050] N=1,441 | 0.003 [0.053] N=1,265 | -0.041 [0.069] N=1,052 | -0.079 [0.090] N=806 |
| Obese: WLZ/WHZ above +3, 0-60 months | -0.034 [0.033] N=1,565 | -0.040 [0.037] N=1,441 | -0.012 [0.039] N=1,265 | -0.028 [0.047] N=1,052 | -0.043 [0.066] N=806 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | -0.049 [0.051] N=1,522 | -0.056 [0.058] N=1,401 | -0.039 [0.063] N=1,229 | -0.119 [0.075] N=1,024 | -0.095 [0.088] N=787 |
| BMI-for-age z score below -2 | -0.042 [0.039] N=1,808 | -0.051 [0.038] N=1,659 | -0.054 [0.045] N=1,453 | -0.041 [0.053] N=1,211 | -0.103 [0.067] N=938 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.069 [0.166] N=242 | -0.042 [0.184] N=220 | 0.059 [0.199] N=193 | -0.162 [0.222] N=165 | -0.183 [0.314] N=132 |
| WHO 2007 BMI obesity above +2, 61+ months | -0.047 [0.121] N=242 | -0.103 [0.132] N=220 | 0.010 [0.156] N=193 | -0.269 [0.171] N=165 | -0.310 [0.251] N=132 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.062 [0.096] N=242 | -0.022 [0.097] N=220 | -0.097 [0.106] N=193 | -0.054 [0.124] N=165 | **-0.315** [0.156] N=132** |

## Mother RDD secondary analysis

*Specification 1 Monthly child-age fixed effects  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.542 | -0.261 [0.189] N=2,274 | -0.295 [0.200] N=2,055 | -0.319 [0.213] N=1,804 | **-0.508** [0.227] N=1,536** | **-0.598** [0.265] N=1,250** |
| WHO weight-for-age z score | -0.064 | -0.061 [0.110] N=2,528 | -0.052 [0.115] N=2,296 | -0.168 [0.120] N=2,010 | -0.147 [0.130] N=1,704 | -0.140 [0.147] N=1,379 |
| WHO weight-for-length/height z score, 0-60 months | 0.366 | 0.065 [0.182] N=1,842 | 0.082 [0.187] N=1,665 | 0.026 [0.211] N=1,454 | 0.142 [0.232] N=1,236 | 0.119 [0.273] N=992 |
| WHO BMI-for-age z score | 0.365 | 0.038 [0.187] N=2,205 | 0.020 [0.197] N=1,994 | -0.013 [0.218] N=1,752 | 0.147 [0.232] N=1,491 | 0.118 [0.256] N=1,213 |
| Stunted: HAZ below -2 | 0.258 P=588 | 0.058 [0.038] N=2,274 | 0.064 [0.040] N=2,055 | 0.065 [0.041] N=1,804 | **0.100** [0.043] N=1,536** | **0.101** [0.051] N=1,250** |
| Underweight: WAZ below -2 | 0.085 P=209 | -0.006 [0.019] N=2,528 | -0.008 [0.021] N=2,296 | 0.010 [0.021] N=2,010 | 0.005 [0.023] N=1,704 | 0.003 [0.023] N=1,379 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=206 | 0.041 [0.029] N=1,842 | 0.037 [0.030] N=1,665 | **0.054* [0.033] N=1,454** | 0.012 [0.034] N=1,236 | 0.003 [0.041] N=992 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.197 P=364 | **0.060* [0.036] N=1,842** | **0.063* [0.037] N=1,665** | 0.054 [0.041] N=1,454 | 0.068 [0.044] N=1,236 | 0.060 [0.050] N=992 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.094 P=180 | 0.026 [0.030] N=1,842 | 0.023 [0.031] N=1,665 | 0.016 [0.034] N=1,454 | 0.019 [0.038] N=1,236 | -0.007 [0.042] N=992 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.358 P=638 | **0.111** [0.044] N=1,785** | **0.103** [0.045] N=1,609** | **0.115** [0.048] N=1,404** | **0.093* [0.050] N=1,192** | **0.096* [0.055] N=957** |
| BMI-for-age z score below -2 | 0.123 P=271 | 0.021 [0.031] N=2,205 | 0.021 [0.033] N=1,994 | 0.024 [0.036] N=1,752 | -0.002 [0.038] N=1,491 | -0.000 [0.043] N=1,213 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.404 P=147 | -0.149 [0.097] N=368 | **-0.244** [0.098] N=332** | **-0.221** [0.101] N=299** | -0.132 [0.105] N=257 | -0.143 [0.114] N=218 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.259 P=91 | -0.017 [0.093] N=368 | -0.112 [0.095] N=332 | -0.120 [0.102] N=299 | -0.007 [0.104] N=257 | -0.023 [0.120] N=218 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=38 | 0.019 [0.062] N=368 | 0.031 [0.065] N=332 | 0.026 [0.066] N=299 | 0.025 [0.068] N=257 | 0.051 [0.080] N=218 |

*Specification 1 Monthly child-age fixed effects  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.077 [0.258] N=1,952 | -0.072 [0.283] N=1,733 | -0.072 [0.319] N=1,482 | -0.483 [0.363] N=1,214 | -0.672 [0.518] N=928 |
| WHO weight-for-age z score | -0.073 [0.148] N=2,176 | -0.059 [0.159] N=1,944 | **-0.307* [0.168] N=1,658** | **-0.343* [0.191] N=1,352** | -0.359 [0.247] N=1,027 |
| WHO weight-for-length/height z score, 0-60 months | -0.099 [0.253] N=1,595 | -0.111 [0.259] N=1,418 | -0.249 [0.307] N=1,207 | -0.074 [0.337] N=989 | -0.272 [0.427] N=745 |
| WHO BMI-for-age z score | -0.287 [0.230] N=1,896 | -0.395 [0.246] N=1,685 | **-0.516* [0.295] N=1,443** | -0.274 [0.332] N=1,182 | -0.547 [0.369] N=904 |
| Stunted: HAZ below -2 | 0.035 [0.058] N=1,952 | 0.041 [0.062] N=1,733 | 0.032 [0.069] N=1,482 | 0.109 [0.076] N=1,214 | 0.127 [0.114] N=928 |
| Underweight: WAZ below -2 | -0.019 [0.028] N=2,176 | -0.021 [0.032] N=1,944 | 0.012 [0.035] N=1,658 | -0.000 [0.041] N=1,352 | -0.014 [0.042] N=1,027 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.098** [0.039] N=1,595** | **0.098** [0.041] N=1,418** | **0.142*** [0.045] N=1,207** | 0.077 [0.047] N=989 | 0.086 [0.058] N=745 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.068 [0.050] N=1,595 | 0.067 [0.053] N=1,418 | 0.058 [0.060] N=1,207 | 0.094 [0.069] N=989 | 0.067 [0.092] N=745 |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Obese: WLZ/WHZ above +3, 0-60 months | 0.033 [0.039] N=1,595 | 0.023 [0.040] N=1,418 | 0.007 [0.045] N=1,207 | 0.016 [0.057] N=989 | -0.078 [0.071] N=745 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.151** [0.069] N=1,547** | **0.146* [0.074] N=1,371** | **0.173** [0.084] N=1,166** | 0.146 [0.101] N=954 | 0.173 [0.128] N=719 |
| BMI-for-age z score below -2 | **0.075** [0.035] N=1,896** | **0.085** [0.039] N=1,685** | **0.096** [0.046] N=1,443** | 0.054 [0.044] N=1,182 | **0.086* [0.050] N=904** |
| WHO 2007 BMI overweight above +1, 61+ months | **-0.259* [0.156] N=302** | **-0.490*** [0.158] N=266** | **-0.502*** [0.179] N=233** | **-0.399* [0.237] N=191** | **-0.587* [0.321] N=152** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.037 [0.145] N=302 | -0.235 [0.154] N=266 | -0.292 [0.182] N=233 | -0.056 [0.201] N=191 | -0.095 [0.304] N=152 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.081 [0.077] N=302 | 0.125 [0.083] N=266 | 0.125 [0.087] N=233 | **0.158* [0.093] N=191** | **0.301** [0.136] N=152** |

*Specification 2 Monthly fixed effects and female  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.542 | -0.260 [0.189] N=2,274 | -0.295 [0.200] N=2,055 | -0.319 [0.213] N=1,804 | **-0.509** [0.227] N=1,536** | **-0.594** [0.266] N=1,250** |
| WHO weight-for-age z score | -0.064 | -0.061 [0.110] N=2,528 | -0.052 [0.115] N=2,296 | -0.168 [0.120] N=2,010 | -0.147 [0.130] N=1,704 | -0.140 [0.147] N=1,379 |
| WHO weight-for-length/height z score, 0-60 months | 0.366 | 0.070 [0.181] N=1,842 | 0.084 [0.187] N=1,665 | 0.027 [0.211] N=1,454 | 0.142 [0.232] N=1,236 | 0.118 [0.271] N=992 |
| WHO BMI-for-age z score | 0.365 | 0.038 [0.187] N=2,205 | 0.021 [0.197] N=1,994 | -0.012 [0.218] N=1,752 | 0.147 [0.232] N=1,491 | 0.116 [0.256] N=1,213 |
| Stunted: HAZ below -2 | 0.258 P=588 | 0.058 [0.038] N=2,274 | 0.064 [0.040] N=2,055 | 0.065 [0.041] N=1,804 | **0.099** [0.044] N=1,536** | **0.102* [0.051] N=1,250** |
| Underweight: WAZ below -2 | 0.085 P=209 | -0.006 [0.019] N=2,528 | -0.008 [0.021] N=2,296 | 0.010 [0.021] N=2,010 | 0.004 [0.023] N=1,704 | 0.003 [0.023] N=1,379 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=206 | 0.039 [0.029] N=1,842 | 0.034 [0.030] N=1,665 | 0.052 [0.033] N=1,454 | 0.011 [0.034] N=1,236 | 0.004 [0.040] N=992 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.197 P=364 | 0.059 [0.036] N=1,842 | 0.061 [0.038] N=1,665 | 0.052 [0.041] N=1,454 | 0.066 [0.045] N=1,236 | 0.060 [0.051] N=992 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.094 P=180 | 0.024 [0.030] N=1,842 | 0.021 [0.031] N=1,665 | 0.014 [0.034] N=1,454 | 0.017 [0.038] N=1,236 | -0.006 [0.043] N=992 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.358 P=638 | **0.109** [0.045] N=1,785** | **0.100** [0.046] N=1,609** | **0.111** [0.048] N=1,404** | **0.092* [0.051] N=1,192** | **0.097* [0.056] N=957** |
| BMI-for-age z score below -2 | 0.123 P=271 | 0.022 [0.031] N=2,205 | 0.021 [0.033] N=1,994 | 0.024 [0.036] N=1,752 | -0.002 [0.038] N=1,491 | 0.001 [0.043] N=1,213 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.404 P=147 | -0.134 [0.098] N=368 | **-0.230** [0.099] N=332** | **-0.211** [0.102] N=299** | -0.130 [0.105] N=257 | -0.137 [0.116] N=218 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.259 P=91 | -0.004 [0.096] N=368 | -0.099 [0.098] N=332 | -0.109 [0.105] N=299 | -0.005 [0.106] N=257 | -0.021 [0.123] N=218 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=38 | 0.009 [0.062] N=368 | 0.022 [0.064] N=332 | 0.015 [0.066] N=299 | 0.020 [0.070] N=257 | 0.040 [0.084] N=218 |

*Specification 2 Monthly fixed effects and female  12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.079 [0.257] N=1,952 | -0.077 [0.281] N=1,733 | -0.077 [0.316] N=1,482 | -0.503 [0.356] N=1,214 | -0.697 [0.510] N=928 |
| WHO weight-for-age z score | -0.073 [0.148] N=2,176 | -0.059 [0.158] N=1,944 | **-0.308* [0.168] N=1,658** | **-0.343* [0.191] N=1,352** | -0.359 [0.247] N=1,027 |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO weight-for-length/height z score, 0-60 months | -0.087 [0.254] N=1,595 | -0.103 [0.260] N=1,418 | -0.246 [0.308] N=1,207 | -0.072 [0.337] N=989 | -0.244 [0.431] N=745 |
| WHO BMI-for-age z score | -0.287 [0.230] N=1,896 | -0.394 [0.247] N=1,685 | **-0.518* [0.295] N=1,443** | -0.275 [0.333] N=1,182 | -0.536 [0.374] N=904 |
| Stunted: HAZ below -2 | 0.035 [0.058] N=1,952 | 0.039 [0.062] N=1,733 | 0.030 [0.068] N=1,482 | 0.107 [0.075] N=1,214 | 0.128 [0.113] N=928 |
| Underweight: WAZ below -2 | -0.019 [0.028] N=2,176 | -0.021 [0.032] N=1,944 | 0.012 [0.035] N=1,658 | -0.001 [0.041] N=1,352 | -0.014 [0.042] N=1,027 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.093** [0.039] N=1,595** | **0.090** [0.042] N=1,418** | **0.134*** [0.045] N=1,207** | 0.067 [0.047] N=989 | 0.073 [0.059] N=745 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.066 [0.050] N=1,595 | 0.062 [0.053] N=1,418 | 0.050 [0.059] N=1,207 | 0.084 [0.068] N=989 | 0.059 [0.091] N=745 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.029 [0.039] N=1,595 | 0.017 [0.040] N=1,418 | -0.002 [0.044] N=1,207 | 0.006 [0.056] N=989 | -0.087 [0.070] N=745 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.144** [0.069] N=1,547** | **0.135* [0.074] N=1,371** | **0.159* [0.084] N=1,166** | 0.132 [0.100] N=954 | 0.161 [0.127] N=719 |
| BMI-for-age z score below -2 | **0.075** [0.035] N=1,896** | **0.083** [0.039] N=1,685** | **0.094** [0.046] N=1,443** | 0.052 [0.045] N=1,182 | 0.084 [0.051] N=904 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.241 [0.157] N=302 | **-0.476*** [0.159] N=266** | **-0.491*** [0.180] N=233** | **-0.399* [0.238] N=191** | **-0.587* [0.323] N=152** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.020 [0.150] N=302 | -0.220 [0.159] N=266 | -0.277 [0.187] N=233 | -0.055 [0.203] N=191 | -0.096 [0.308] N=152 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.074 [0.078] N=302 | 0.119 [0.083] N=266 | 0.116 [0.086] N=233 | **0.157* [0.092] N=191** | **0.291** [0.144] N=152** |

*Specification 3 Monthly fixed effects female and household income groups  No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.542 | -0.259 [0.185] N=2,274 | -0.286 [0.195] N=2,055 | -0.300 [0.211] N=1,804 | **-0.488** [0.221] N=1,536** | **-0.573** [0.255] N=1,250** |
| WHO weight-for-age z score | -0.064 | -0.058 [0.108] N=2,528 | -0.046 [0.113] N=2,296 | -0.154 [0.119] N=2,010 | -0.143 [0.130] N=1,704 | -0.136 [0.148] N=1,379 |
| WHO weight-for-length/height z score, 0-60 months | 0.366 | 0.079 [0.182] N=1,842 | 0.090 [0.188] N=1,665 | 0.040 [0.213] N=1,454 | 0.142 [0.233] N=1,236 | 0.118 [0.271] N=992 |
| WHO BMI-for-age z score | 0.365 | 0.039 [0.186] N=2,205 | 0.020 [0.196] N=1,994 | -0.009 [0.217] N=1,752 | 0.144 [0.230] N=1,491 | 0.112 [0.257] N=1,213 |
| Stunted: HAZ below -2 | 0.258 P=588 | 0.058 [0.038] N=2,274 | 0.061 [0.039] N=2,055 | 0.058 [0.041] N=1,804 | **0.091** [0.042] N=1,536** | **0.095* [0.049] N=1,250** |
| Underweight: WAZ below -2 | 0.085 P=209 | -0.006 [0.019] N=2,528 | -0.008 [0.021] N=2,296 | 0.007 [0.021] N=2,010 | 0.003 [0.023] N=1,704 | 0.002 [0.022] N=1,379 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=206 | 0.038 [0.030] N=1,842 | 0.032 [0.031] N=1,665 | 0.049 [0.033] N=1,454 | 0.010 [0.035] N=1,236 | 0.003 [0.040] N=992 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.197 P=364 | **0.062* [0.036] N=1,842** | 0.062 [0.038] N=1,665 | 0.052 [0.042] N=1,454 | 0.064 [0.045] N=1,236 | 0.060 [0.051] N=992 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.094 P=180 | 0.026 [0.031] N=1,842 | 0.022 [0.031] N=1,665 | 0.014 [0.034] N=1,454 | 0.016 [0.039] N=1,236 | -0.006 [0.044] N=992 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.358 P=638 | **0.111** [0.044] N=1,785** | **0.098** [0.046] N=1,609** | **0.106** [0.048] N=1,404** | **0.085* [0.050] N=1,192** | **0.093* [0.054] N=957** |
| BMI-for-age z score below -2 | 0.123 P=271 | 0.022 [0.031] N=2,205 | 0.021 [0.033] N=1,994 | 0.022 [0.036] N=1,752 | -0.004 [0.038] N=1,491 | -0.001 [0.044] N=1,213 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.404 P=147 | -0.148 [0.098] N=368 | **-0.242** [0.100] N=332** | **-0.230** [0.106] N=299** | -0.136 [0.109] N=257 | -0.176 [0.129] N=218 |

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO 2007 BMI obesity above +2, 61+ months | 0.259 P=91 | -0.020 [0.094] N=368 | -0.110 [0.098] N=332 | -0.124 [0.106] N=299 | -0.023 [0.107] N=257 | -0.054 [0.127] N=218 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=38 | 0.004 [0.063] N=368 | 0.017 [0.063] N=332 | 0.013 [0.065] N=299 | 0.008 [0.068] N=257 | 0.026 [0.078] N=218 |

*Specification 3 Monthly fixed effects female and household income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.068 [0.253] N=1,952 | -0.059 [0.275] N=1,733 | -0.038 [0.311] N=1,482 | -0.458 [0.347] N=1,214 | -0.631 [0.493] N=928 |
| WHO weight-for-age z score | -0.070 [0.147] N=2,176 | -0.054 [0.159] N=1,944 | **-0.283* [0.170] N=1,658** | **-0.338* [0.192] N=1,352** | -0.349 [0.245] N=1,027 |
| WHO weight-for-length/height z score, 0-60 months | -0.096 [0.254] N=1,595 | -0.111 [0.262] N=1,418 | -0.246 [0.309] N=1,207 | -0.089 [0.336] N=989 | -0.253 [0.426] N=745 |
| WHO BMI-for-age z score | -0.300 [0.230] N=1,896 | -0.404 [0.247] N=1,685 | **-0.527* [0.293] N=1,443** | -0.299 [0.332] N=1,182 | -0.573 [0.373] N=904 |
| Stunted: HAZ below -2 | 0.032 [0.057] N=1,952 | 0.035 [0.061] N=1,733 | 0.020 [0.067] N=1,482 | 0.095 [0.074] N=1,214 | 0.111 [0.108] N=928 |
| Underweight: WAZ below -2 | -0.020 [0.028] N=2,176 | -0.023 [0.031] N=1,944 | 0.004 [0.035] N=1,658 | -0.005 [0.041] N=1,352 | -0.020 [0.042] N=1,027 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.093** [0.040] N=1,595** | **0.089** [0.042] N=1,418** | **0.130*** [0.046] N=1,207** | 0.064 [0.048] N=989 | 0.073 [0.060] N=745 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.067 [0.051] N=1,595 | 0.063 [0.054] N=1,418 | 0.047 [0.061] N=1,207 | 0.078 [0.070] N=989 | 0.058 [0.093] N=745 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.029 [0.039] N=1,595 | 0.015 [0.040] N=1,418 | -0.003 [0.045] N=1,207 | 0.003 [0.057] N=989 | -0.089 [0.071] N=745 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.145** [0.069] N=1,547** | **0.132* [0.075] N=1,371** | **0.148* [0.085] N=1,166** | 0.122 [0.100] N=954 | 0.158 [0.125] N=719 |
| BMI-for-age z score below -2 | **0.075** [0.035] N=1,896** | **0.082** [0.039] N=1,685** | **0.089* [0.045] N=1,443** | 0.048 [0.046] N=1,182 | 0.078 [0.052] N=904 |
| WHO 2007 BMI overweight above +1, 61+ months | -0.253 [0.155] N=302 | **-0.488*** [0.155] N=266** | **-0.527*** [0.174] N=233** | **-0.410* [0.234] N=191** | **-0.808** [0.306] N=152** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.033 [0.147] N=302 | -0.232 [0.156] N=266 | -0.305 [0.184] N=233 | -0.084 [0.196] N=191 | -0.281 [0.311] N=152 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.071 [0.078] N=302 | 0.115 [0.083] N=266 | 0.111 [0.086] N=233 | 0.130 [0.094] N=191 | 0.240 [0.167] N=152 |

*Specification 4 Monthly fixed effects female and per-person income groups No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.542 | -0.256 [0.184] N=2,274 | -0.280 [0.195] N=2,055 | -0.296 [0.210] N=1,804 | **-0.483** [0.219] N=1,536** | **-0.578** [0.253] N=1,250** |
| WHO weight-for-age z score | -0.064 | -0.057 [0.107] N=2,528 | -0.042 [0.113] N=2,296 | -0.148 [0.119] N=2,010 | -0.124 [0.128] N=1,704 | -0.131 [0.144] N=1,379 |
| WHO weight-for-length/height z score, 0-60 months | 0.366 | 0.083 [0.181] N=1,842 | 0.096 [0.186] N=1,665 | 0.042 [0.211] N=1,454 | 0.157 [0.230] N=1,236 | 0.140 [0.270] N=992 |
| WHO BMI-for-age z score | 0.365 | 0.039 [0.186] N=2,205 | 0.023 [0.197] N=1,994 | -0.005 [0.217] N=1,752 | 0.158 [0.229] N=1,491 | 0.122 [0.254] N=1,213 |
| Stunted: HAZ below -2 | 0.258 P=588 | 0.058 [0.037] N=2,274 | 0.061 [0.039] N=2,055 | 0.060 [0.041] N=1,804 | **0.093** [0.042] N=1,536** | **0.096* [0.049] N=1,250** |
| Underweight: WAZ below -2 | 0.085 P=209 | -0.006 [0.019] N=2,528 | -0.009 [0.020] N=2,296 | 0.008 [0.021] N=2,010 | 0.002 [0.023] N=1,704 | 0.002 [0.023] N=1,379 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.111 P=206 | 0.037 [0.029] N=1,842 | 0.033 [0.030] N=1,665 | 0.051 [0.033] N=1,454 | 0.009 [0.034] N=1,236 | 0.000 [0.040] N=992 |

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.197 P=364 | **0.060* [0.036] N=1,842** | 0.062 [0.037] N=1,665 | 0.054 [0.041] N=1,454 | 0.067 [0.044] N=1,236 | 0.062 [0.050] N=992 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.094 P=180 | 0.028 [0.030] N=1,842 | 0.025 [0.030] N=1,665 | 0.017 [0.033] N=1,454 | 0.021 [0.038] N=1,236 | -0.002 [0.043] N=992 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.358 P=638 | **0.108** [0.044] N=1,785** | **0.096** [0.046] N=1,609** | **0.106** [0.048] N=1,404** | **0.085* [0.050] N=1,192** | **0.092* [0.054] N=957** |
| BMI-for-age z score below -2 | 0.123 P=271 | 0.021 [0.031] N=2,205 | 0.019 [0.034] N=1,994 | 0.021 [0.036] N=1,752 | -0.006 [0.038] N=1,491 | -0.004 [0.043] N=1,213 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.404 P=147 | -0.143 [0.098] N=368 | **-0.233** [0.101] N=332** | **-0.208* [0.107] N=299** | -0.120 [0.111] N=257 | -0.147 [0.127] N=218 |
| WHO 2007 BMI obesity above +2, 61+ months | 0.259 P=91 | -0.018 [0.097] N=368 | -0.110 [0.100] N=332 | -0.119 [0.108] N=299 | -0.020 [0.109] N=257 | -0.057 [0.125] N=218 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.109 P=38 | 0.006 [0.062] N=368 | 0.019 [0.063] N=332 | 0.007 [0.064] N=299 | 0.012 [0.067] N=257 | 0.009 [0.076] N=218 |

*Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.063 [0.252] N=1,952 | -0.044 [0.275] N=1,733 | -0.011 [0.312] N=1,482 | -0.422 [0.347] N=1,214 | -0.651 [0.484] N=928 |
| WHO weight-for-age z score | -0.056 [0.144] N=2,176 | -0.032 [0.154] N=1,944 | -0.255 [0.166] N=1,658 | -0.274 [0.184] N=1,352 | -0.314 [0.231] N=1,027 |
| WHO weight-for-length/height z score, 0-60 months | -0.084 [0.253] N=1,595 | -0.107 [0.258] N=1,418 | -0.252 [0.301] N=1,207 | -0.085 [0.329] N=989 | -0.230 [0.425] N=745 |
| WHO BMI-for-age z score | -0.298 [0.230] N=1,896 | -0.405 [0.247] N=1,685 | **-0.529* [0.293] N=1,443** | -0.278 [0.325] N=1,182 | -0.537 [0.371] N=904 |
| Stunted: HAZ below -2 | 0.033 [0.056] N=1,952 | 0.035 [0.060] N=1,733 | 0.020 [0.066] N=1,482 | 0.093 [0.073] N=1,214 | 0.119 [0.109] N=928 |
| Underweight: WAZ below -2 | -0.022 [0.027] N=2,176 | -0.027 [0.031] N=1,944 | 0.002 [0.034] N=1,658 | -0.012 [0.040] N=1,352 | -0.022 [0.041] N=1,027 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.092** [0.040] N=1,595** | **0.089** [0.042] N=1,418** | **0.135*** [0.045] N=1,207** | 0.068 [0.047] N=989 | 0.075 [0.058] N=745 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.065 [0.050] N=1,595 | 0.060 [0.052] N=1,418 | 0.046 [0.059] N=1,207 | 0.075 [0.069] N=989 | 0.060 [0.093] N=745 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.031 [0.039] N=1,595 | 0.018 [0.040] N=1,418 | -0.001 [0.045] N=1,207 | 0.006 [0.057] N=989 | -0.085 [0.070] N=745 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.142** [0.068] N=1,547** | **0.128* [0.074] N=1,371** | **0.146* [0.084] N=1,166** | 0.116 [0.098] N=954 | 0.162 [0.121] N=719 |
| BMI-for-age z score below -2 | **0.076** [0.035] N=1,896** | **0.082** [0.040] N=1,685** | **0.091** [0.046] N=1,443** | 0.049 [0.046] N=1,182 | 0.080 [0.051] N=904 |
| WHO 2007 BMI overweight above +1, 61+ months | **-0.257* [0.150] N=302** | **-0.478*** [0.150] N=266** | **-0.482*** [0.170] N=233** | -0.381 [0.230] N=191 | **-0.712** [0.289] N=152** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.027 [0.145] N=302 | -0.226 [0.153] N=266 | -0.284 [0.181] N=233 | -0.071 [0.198] N=191 | -0.225 [0.313] N=152 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.079 [0.078] N=302 | 0.120 [0.083] N=266 | 0.105 [0.083] N=233 | 0.134 [0.088] N=191 | 0.181 [0.142] N=152 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.553 | -0.277 [0.188] N=2,131 | -0.284 [0.202] N=1,926 | -0.287 [0.215] N=1,694 | **-0.479** [0.229] N=1,438** | **-0.609** [0.263] N=1,173** |

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO weight-for-age z score | -0.061 | -0.049 [0.111] N=2,361 | -0.022 [0.117] N=2,143 | -0.142 [0.122] N=1,880 | -0.131 [0.132] N=1,592 | -0.163 [0.148] N=1,293 |
| WHO weight-for-length/height z score, 0-60 months | 0.368 | 0.116 [0.192] N=1,719 | 0.144 [0.199] N=1,554 | 0.055 [0.222] N=1,358 | 0.154 [0.243] N=1,153 | 0.094 [0.279] N=928 |
| WHO BMI-for-age z score | 0.377 | 0.026 [0.195] N=2,065 | 0.011 [0.206] N=1,868 | -0.067 [0.229] N=1,644 | 0.071 [0.245] N=1,397 | 0.005 [0.270] N=1,140 |
| Stunted: HAZ below -2 | 0.258 P=550 | 0.060 [0.038] N=2,131 | 0.064 [0.040] N=1,926 | 0.059 [0.041] N=1,694 | **0.093** [0.042] N=1,438** | **0.101** [0.048] N=1,173** |
| Underweight: WAZ below -2 | 0.086 P=197 | -0.011 [0.020] N=2,361 | -0.014 [0.022] N=2,143 | 0.000 [0.022] N=1,880 | -0.000 [0.024] N=1,592 | 0.006 [0.025] N=1,293 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.112 P=193 | 0.042 [0.030] N=1,719 | 0.035 [0.031] N=1,554 | 0.054 [0.033] N=1,358 | 0.017 [0.035] N=1,153 | 0.016 [0.041] N=928 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.198 P=341 | 0.059 [0.037] N=1,719 | 0.062 [0.038] N=1,554 | 0.048 [0.041] N=1,358 | 0.059 [0.044] N=1,153 | 0.048 [0.050] N=928 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.096 P=172 | 0.024 [0.031] N=1,719 | 0.021 [0.031] N=1,554 | 0.012 [0.033] N=1,358 | 0.019 [0.038] N=1,153 | -0.001 [0.042] N=928 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.359 P=597 | **0.116** [0.047] N=1,667** | **0.104** [0.049] N=1,503** | **0.109** [0.050] N=1,313** | 0.087 [0.053] N=1,112 | **0.103* [0.057] N=895** |
| BMI-for-age z score below -2 | 0.121 P=247 | 0.032 [0.030] N=2,065 | 0.032 [0.032] N=1,868 | 0.037 [0.035] N=1,644 | 0.017 [0.036] N=1,397 | 0.023 [0.040] N=1,140 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.410 P=142 | **-0.179* [0.100] N=352** | **-0.294*** [0.102] N=318** | **-0.274** [0.107] N=287** | **-0.207* [0.115] N=246** | **-0.226* [0.126] N=209** |
| WHO 2007 BMI obesity above +2, 61+ months | 0.264 P=88 | -0.037 [0.102] N=352 | -0.142 [0.104] N=318 | -0.159 [0.112] N=287 | -0.067 [0.113] N=246 | -0.069 [0.134] N=209 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.101 P=33 | 0.036 [0.063] N=352 | 0.048 [0.065] N=318 | 0.041 [0.068] N=287 | 0.055 [0.072] N=246 | 0.090 [0.084] N=209 |

*Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.079 [0.264] N=1,828 | -0.035 [0.295] N=1,623 | -0.004 [0.337] N=1,391 | -0.420 [0.381] N=1,135 | -0.697 [0.527] N=870 |
| WHO weight-for-age z score | -0.064 [0.149] N=2,032 | -0.017 [0.161] N=1,814 | **-0.284* [0.169] N=1,551** | -0.324 [0.196] N=1,263 | -0.355 [0.257] N=964 |
| WHO weight-for-length/height z score, 0-60 months | -0.034 [0.267] N=1,487 | -0.027 [0.280] N=1,322 | -0.241 [0.331] N=1,126 | -0.093 [0.354] N=921 | -0.244 [0.466] N=696 |
| WHO BMI-for-age z score | -0.266 [0.235] N=1,773 | -0.352 [0.256] N=1,576 | **-0.536* [0.311] N=1,352** | -0.322 [0.341] N=1,105 | -0.580 [0.392] N=848 |
| Stunted: HAZ below -2 | 0.030 [0.059] N=1,828 | 0.034 [0.065] N=1,623 | 0.011 [0.072] N=1,391 | 0.077 [0.077] N=1,135 | 0.101 [0.108] N=870 |
| Underweight: WAZ below -2 | -0.030 [0.029] N=2,032 | -0.035 [0.033] N=1,814 | -0.008 [0.036] N=1,551 | -0.013 [0.044] N=1,263 | -0.024 [0.046] N=964 |
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.095** [0.040] N=1,487** | **0.086** [0.043] N=1,322** | **0.132*** [0.047] N=1,126** | 0.070 [0.051] N=921 | 0.074 [0.061] N=696 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.066 [0.052] N=1,487 | 0.063 [0.055] N=1,322 | 0.041 [0.061] N=1,126 | 0.066 [0.071] N=921 | 0.046 [0.098] N=696 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.023 [0.041] N=1,487 | 0.010 [0.042] N=1,322 | -0.014 [0.047] N=1,126 | -0.001 [0.059] N=921 | -0.084 [0.077] N=696 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.150** [0.074] N=1,443** | **0.135* [0.081] N=1,279** | 0.143 [0.090] N=1,089 | 0.093 [0.110] N=888 | 0.130 [0.131] N=671 |
| BMI-for-age z score below -2 | **0.081** [0.035] N=1,773** | **0.088** [0.041] N=1,576** | **0.101** [0.047] N=1,352** | 0.072 [0.047] N=1,105 | **0.101* [0.053] N=848** |
| WHO 2007 BMI overweight above +1, 61+ months | **-0.295* [0.158] N=289** | **-0.544*** [0.163] N=255** | **-0.540*** [0.184] N=224** | **-0.470* [0.240] N=183** | **-0.708** [0.329] N=146** |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO 2007 BMI obesity above +2, 61+ months | -0.050 [0.155] N=289 | -0.243 [0.171] N=255 | -0.299 [0.197] N=224 | -0.076 [0.214] N=183 | -0.088 [0.324] N=146 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.085 [0.081] N=289 | 0.130 [0.084] N=255 | 0.122 [0.087] N=224 | **0.179** [0.087] N=183** | **0.324** [0.143] N=146** |

*Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole*

| Outcome | Mean | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.553 | -0.280 [0.190] N=2,131 | -0.289 [0.204] N=1,926 | -0.293 [0.217] N=1,694 | **-0.479** [0.232] N=1,438** | **-0.607** [0.264] N=1,173** |
| WHO weight-for-age z score | -0.061 | -0.048 [0.111] N=2,361 | -0.025 [0.118] N=2,143 | -0.148 [0.123] N=1,880 | -0.134 [0.134] N=1,592 | -0.164 [0.151] N=1,293 |
| WHO weight-for-length/height z score, 0-60 months | 0.368 | 0.124 [0.192] N=1,719 | 0.148 [0.200] N=1,554 | 0.060 [0.222] N=1,358 | 0.169 [0.240] N=1,153 | 0.112 [0.280] N=928 |
| WHO BMI-for-age z score | 0.377 | 0.028 [0.195] N=2,065 | 0.013 [0.206] N=1,868 | -0.063 [0.228] N=1,644 | 0.074 [0.245] N=1,397 | 0.010 [0.270] N=1,140 |
| Stunted: HAZ below -2 | 0.258 P=550 | 0.058 [0.038] N=2,131 | 0.062 [0.040] N=1,926 | 0.057 [0.041] N=1,694 | **0.091** [0.042] N=1,438** | **0.098** [0.048] N=1,173** |
| Underweight: WAZ below -2 | 0.086 P=197 | -0.012 [0.020] N=2,361 | -0.014 [0.022] N=2,143 | 0.002 [0.022] N=1,880 | -0.000 [0.024] N=1,592 | 0.005 [0.025] N=1,293 |
| Wasted: WLZ/WHZ below -2, 0-60 months | 0.112 P=193 | 0.040 [0.030] N=1,719 | 0.033 [0.031] N=1,554 | 0.053 [0.033] N=1,358 | 0.012 [0.035] N=1,153 | 0.010 [0.040] N=928 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.198 P=341 | 0.059 [0.036] N=1,719 | 0.061 [0.038] N=1,554 | 0.047 [0.041] N=1,358 | 0.058 [0.044] N=1,153 | 0.048 [0.050] N=928 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.096 P=172 | 0.024 [0.030] N=1,719 | 0.021 [0.030] N=1,554 | 0.011 [0.033] N=1,358 | 0.018 [0.037] N=1,153 | -0.002 [0.042] N=928 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | 0.359 P=597 | **0.112** [0.046] N=1,667** | **0.100** [0.048] N=1,503** | **0.104** [0.050] N=1,313** | 0.080 [0.052] N=1,112 | **0.093* [0.055] N=895** |
| BMI-for-age z score below -2 | 0.121 P=247 | 0.031 [0.030] N=2,065 | 0.031 [0.032] N=1,868 | 0.035 [0.035] N=1,644 | 0.015 [0.036] N=1,397 | 0.020 [0.040] N=1,140 |
| WHO 2007 BMI overweight above +1, 61+ months | 0.410 P=142 | **-0.183* [0.101] N=352** | **-0.304*** [0.101] N=318** | **-0.288*** [0.105] N=287** | **-0.219* [0.111] N=246** | **-0.231* [0.122] N=209** |
| WHO 2007 BMI obesity above +2, 61+ months | 0.264 P=88 | -0.034 [0.101] N=352 | -0.147 [0.104] N=318 | -0.165 [0.110] N=287 | -0.078 [0.113] N=246 | -0.081 [0.132] N=209 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.101 P=33 | 0.031 [0.062] N=352 | 0.049 [0.064] N=318 | 0.038 [0.065] N=287 | 0.054 [0.068] N=246 | 0.087 [0.080] N=209 |

*Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole*

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| WHO length/height-for-age z score | -0.094 [0.264] N=1,828 | -0.050 [0.293] N=1,623 | -0.021 [0.334] N=1,391 | -0.469 [0.380] N=1,135 | -0.752 [0.520] N=870 |
| WHO weight-for-age z score | -0.062 [0.148] N=2,032 | -0.021 [0.161] N=1,814 | **-0.294* [0.169] N=1,551** | **-0.333* [0.195] N=1,263** | -0.403 [0.255] N=964 |
| WHO weight-for-length/height z score, 0-60 months | -0.035 [0.266] N=1,487 | -0.030 [0.276] N=1,322 | -0.240 [0.327] N=1,126 | -0.063 [0.354] N=921 | -0.223 [0.456] N=696 |
| WHO BMI-for-age z score | -0.260 [0.235] N=1,773 | -0.349 [0.254] N=1,576 | **-0.537* [0.307] N=1,352** | -0.301 [0.339] N=1,105 | -0.565 [0.390] N=848 |
| Stunted: HAZ below -2 | 0.031 [0.059] N=1,828 | 0.035 [0.064] N=1,623 | 0.013 [0.071] N=1,391 | 0.085 [0.076] N=1,135 | 0.114 [0.106] N=870 |
| Underweight: WAZ below -2 | -0.030 [0.029] N=2,032 | -0.035 [0.033] N=1,814 | -0.006 [0.035] N=1,551 | -0.011 [0.043] N=1,263 | -0.014 [0.045] N=964 |

| Outcome | 8 years | 7 years | 6 years | 5 years | 4 years |
|---|---|---|---|---|---|
| Wasted: WLZ/WHZ below -2, 0-60 months | **0.095** [0.040] N=1,487** | **0.087** [0.043] N=1,322** | **0.134*** [0.046] N=1,126** | 0.066 [0.050] N=921 | 0.073 [0.059] N=696 |
| Overweight: WLZ/WHZ above +2, 0-60 months | 0.066 [0.051] N=1,487 | 0.063 [0.053] N=1,322 | 0.040 [0.060] N=1,126 | 0.070 [0.070] N=921 | 0.048 [0.096] N=696 |
| Obese: WLZ/WHZ above +3, 0-60 months | 0.023 [0.041] N=1,487 | 0.010 [0.042] N=1,322 | -0.013 [0.046] N=1,126 | -0.001 [0.059] N=921 | -0.084 [0.075] N=696 |
| CIAF: any stunting, underweight, or wasting, 0-60 months | **0.149** [0.073] N=1,443** | **0.134* [0.080] N=1,279** | 0.144 [0.090] N=1,089 | 0.098 [0.109] N=888 | 0.136 [0.130] N=671 |
| BMI-for-age z score below -2 | **0.081** [0.035] N=1,773** | **0.088** [0.041] N=1,576** | **0.102** [0.046] N=1,352** | 0.070 [0.046] N=1,105 | **0.102* [0.052] N=848** |
| WHO 2007 BMI overweight above +1, 61+ months | **-0.276* [0.158] N=289** | **-0.529*** [0.160] N=255** | **-0.545*** [0.181] N=224** | **-0.471** [0.234] N=183** | **-0.699** [0.317] N=146** |
| WHO 2007 BMI obesity above +2, 61+ months | -0.028 [0.153] N=289 | -0.237 [0.167] N=255 | -0.297 [0.193] N=224 | -0.088 [0.211] N=183 | -0.116 [0.316] N=146 |
| WHO 2007 BMI thinness below -2, 61+ months | 0.079 [0.079] N=289 | 0.130 [0.082] N=255 | 0.114 [0.086] N=224 | **0.161* [0.085] N=183** | **0.285** [0.141] N=146** |

*Notes:* Cells report the OLS policy coefficient, birth-month-clustered standard error in brackets, and N. Exact child-month fixed effects for analysis months 0 to 71 are included but not displayed. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add the categorical number of aid types (0 as reference) and any aid. Aid and income controls are not combined. Aid specifications exclude households with a blank or non-yes/no answer on any of the six main aid questions. Columns use symmetric 8- to 4-year bandwidths around the January 1987 cutoff. Every regression includes reform eligibility and split linear trends. Donut estimates exclude running-variable months -12 through +11 and show only the policy estimate, standard error, and N. Regressions use FAKTOR_FERT child survey weights. The sample retains one latest-born eligible child per household. Significant policy coefficients are bold. P in the mean column is the unweighted number of positive outcomes in the no-donut 8-year sample. *** p<0.01, ** p<0.05, * p<0.10.